In-Session Activity: Subnetting

NOTE: All the calculations, the drawing, and the answers are done by me. I used Google to do research so that I understand what's going on (the Subnetting calculator made no sense).

The Scenario was taken from the Canvas assignment.

Scenario

You are tasked with designing a subnetted network for CyberSecure Ltd. with **three departments**. Each department has different IP needs based on the number of devices and users they support.

You will use the base network **10.0.0.0/24** to divide into subnets that best fit each department's requirements.

Subnet Design Plan

Let's take three hypothetical departments of CyberSecure Ltd.: Sales (because every single company needs to generate sales), Marketing (to promote CyberSecure), and HR (because every company needs HR in order to function).

Here's the full rundown, along with the calculations:

Dept.	Network Address	Address Needed	Subnet Size (2^n)	Subnet Mask (32-n)	CIDR	Usable IP Range	Available Hosts Per subnet ((2^n) - 2)	Broadcast Address
Sales	10.0.0.0	100	128	25	/25	10.0.0.1-10.0.0.126	126	10.0.0.127
Marketing	10.0.0.128	50	64	26	/26	10.0.0.129-10.0.0.190	62	10.0.0.191
HR	10.0.0.192	25	32	27	/27	10.0.0.193-10.0.0.222	30	10.0.0.223

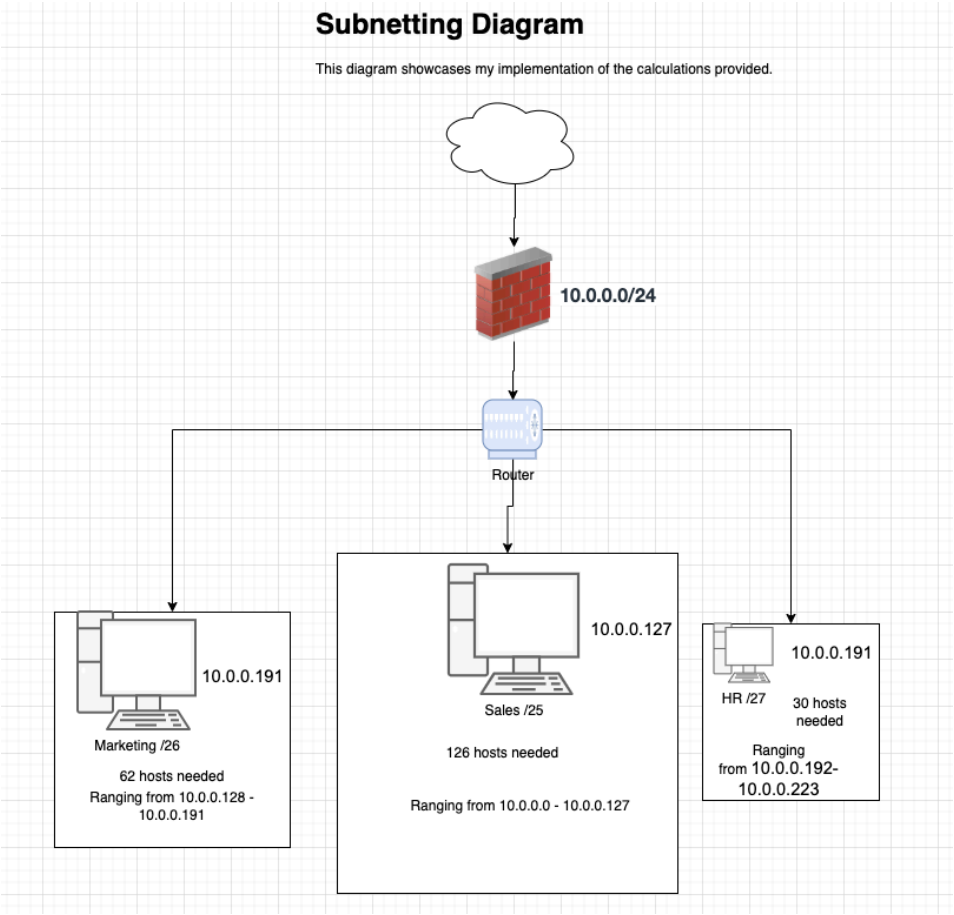
According to Google, "every subnet size must be a power of 2". Google also stated that to find the subnet mask, we take the number of bits for an IPv4 address (32 bits) and subtract *n*.

As soon as the CIDR was found, the cheat sheet (provided on Canvas) explained the rest:

Subnets			
CIDR	Subnet Mask	# of Addresses	Wildcard
/0	0.0.0.0	4,294,967,296	255.255.255.255
/1	128.0.0.0	2,147,483,648	127.255.255.255
/2	192.0.0.0	1,073,741,824	63.255.255.255
/3	224.0.0.0	536,870,912	31.255.255.255
/4	240.0.0.0	268,435,456	15.255.255.255
/5	248.0.0.0	134,217,728	7.255.255.255
/6	252.0.0.0	67,108,864	3.255.255.255
/7	254.0.0.0	33,554,432	1.255.255.255
/8	255.0.0.0	16,777,216	0.255.255.255
/9	255.128.0.0	8,388,608	0.127.255.255
/10	255.192.0.0	4,194,304	0.63.255.255
/11	255.224.0.0	2,097,152	0.31.255.255
/12	255.240.0.0	1,048,576	0.15.255.255
/13	255.248.0.0	524,288	0.7.255.255
/14	255.252.0.0	262,144	0.3.255.255
/15	255.254.0.0	131,072	0.1.255.255
/16	255.255.0.0	65,536	0.0.255.255
/17	255.255.128.0	32,768	0.0.127.255
/18	255.255.192.0	16,384	0.0.63.255
/19	255.255.224.0	8,192	0.0.31.255
/20	255.255.240.0	4,096	0.0.15.255
/21	255.255.248.0	2,048	0.0.7.255
/22	255.255.252.0	1,024	0.0.3.255
/23	255.255.254.0	512	0.0.1.255
/24	255.255.255.0	256	0.0.0.255
/25	255.255.255.128	128	0.0.0.127
/26	255.255.255.192	64	0.0.0.63
/27	255.255.255.224	32	0.0.0.31
/28	255.255.255.240	16	0.0.0.15
/29	255.255.255.248	8	0.0.0.7
/30	255.255.255.252	4	0.0.0.3
/31	255.255.255.254	2	0.0.0.1
/32	255.255.255.255	1	0.0.0.0

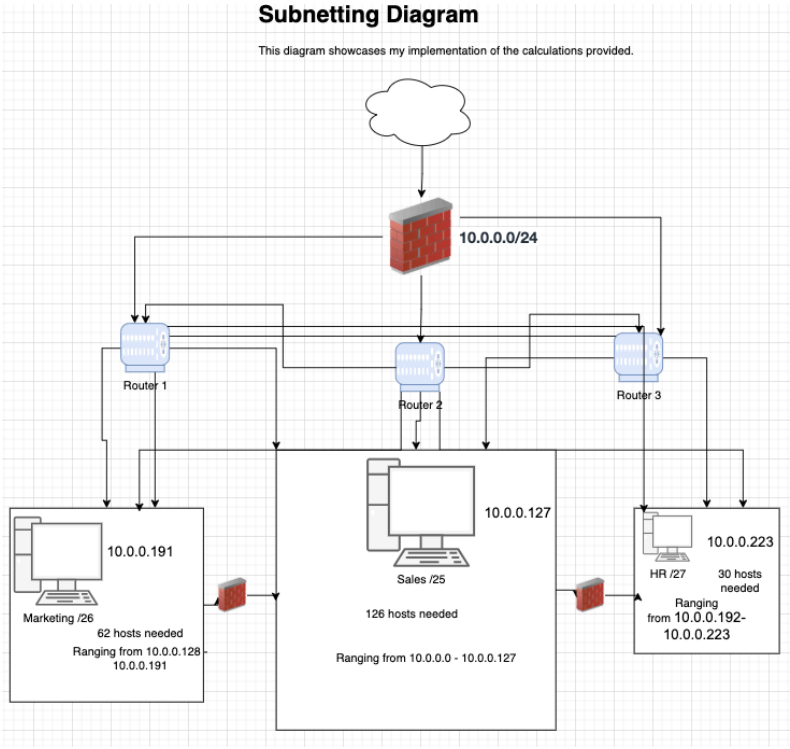
Subnet Design

The diagram provides a graphical explanation of my calculations:



Reflection Questions

1. I used /25, /26, /27 out of respect for the departments (Sales, HR, and Marketing). If I used /26 for each and every one of those departments, it would cause **IP Fragmentation** (breaking data packets into smaller ones so they can pass through a network link with a smaller size limit), creating problems for the other, smaller departments. Giving the largest block to the department with the most hosts first solved this problem.
2. Routing restrictions might be helpful between Sales and HR. I wouldn't want HR to access the Sales computers - but at the same time, I don't want Sales to have unrestricted access to the HR's servers.
3. The Canvas assignment told me to create a diagram that showcases a subnetted network for a company with three departments. The design shown above represents that. However, it's just a rudimentary plan - it showcases a router connecting to 3 departments. **All 3 departments rely on that router.** Router goes down, company goes down.

Additionally, as mentioned before, all three departments rely on a single router. There is no firewall or ACL in place. Without it, an attacker might perform lateral movement and insert a piece of malware that could easily spread to other departments.

This diagram should resolve the issue: